

Study of glycated human serum albumin in non-enzymatic glycation process based on MIR/NIR spectroscopy

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ABSTRACT

Non-enzymatic glycation of human serum albumin (HSA) is implicated in diabetes mellitus, its complications, and neurodegenerative diseases. This complex process yields diverse products across various stages, yet traditional assays lack the capability to characterize early and intermediate glycation phases effectively. Infrared spectroscopy, encompassing near-infrared (NIR) and mid-infrared (MIR) regions, offers insights into molecular vibrations and has gained traction in studying protein-molecule interactions. Our study employed NIR and MIR spectroscopy to monitor the glycation of HSA induced by 50 mM glucose over five weeks, establishing quantitative models for glycated HSA. NIR analysis revealed that HSA produced the highest amount of fructosamine at 3 weeks, while five characteristic peaks 4768 cm^{-1} , 5644 cm^{-1} , 5982 cm^{-1} , 7012 cm^{-1} , 7143 cm^{-1} were found. Meanwhile, MIR spectroscopy further revealed the peaks 675 cm^{-1} , 1517 cm^{-1} , 1685 cm^{-1} , 1792 cm^{-1} , and 1840 cm^{-1} , which reflected the degree of glycation of HSA. A robust quantitative model, integrating NIR and MIR data, demonstrated high predictive accuracy ($R^2\text{c} = 0.9994$, $R^2\text{p} = 0.9524$, RMSEP = 1.59 mmol/L) and reliability (RPD = 3.35). This research not only elucidates HSA glycation levels but also pioneers a novel quantification methodology for glycated HSA.

1. Introduction

Diabetes prevalence continues to rise at an alarming rate worldwide. According to the International Diabetes Federation (IDF) Global Diabetes Map (10th Edition, 2021), the global diabetes prevalence among individuals aged 20–79 years reached 10.5 % (representing 536.6 million people) in 2021, with projections indicating a significant increase to 12.2 % (affecting 783.2 million people) by 2045 [1]. Human Serum Albumin (HSA), a 585-amino acid protein constituting approximately half of the serum protein content in human blood plasma, serves as the primary circulating protein with multifaceted physiological functions [2,3]. These critical functions encompass the regulation of blood osmotic pressure and pH balance, facilitation of lipid metabolism,

transportation of endogenous and exogenous substances, and provision of binding sites for numerous therapeutic agents. Under conditions of elevated blood glucose levels, HSA undergoes non-enzymatic glycation, forming glycated HSA (gHSA), which has emerged as a significant biomarker for diabetes mellitus and its associated complications. Unlike conventional biomarkers, gHSA reflects glycemic fluctuations over a 2–3 week period, thereby providing a valuable window for monitoring therapeutic interventions, lifestyle modifications, and early glycemic control [4]. The clinical significance of gHSA extends beyond its role as an alternative glycemic marker when HbA1c measurements are unreliable. Substantial evidence positions gHSA as an independent risk factor for the progression of diabetic complications and insulin resistance [5]. Particularly in gestational diabetes mellitus (GDM) management,

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emerging studies suggest gHSA's superior monitoring capabilities compared to traditional HbA1c measurements [6]. This diagnostic advantage is particularly crucial given that diabetes-related complications, rather than the disease itself, represent the primary mortality risk. Early detection and intervention through gHSA monitoring can significantly delay the onset of both microvascular and macrovascular complications, including cardiovascular diseases [7].

The pathophysiological implications of elevated gHSA levels are profound, with demonstrated capacity to induce irreversible damage to multiple organ systems, particularly the cardiovascular system, coronary arteries, ocular structures, thyroid function, and renal apparatus. The molecular mechanisms underlying these effects involve the stimulation of inflammatory cytokine production in renal epithelial and mesangial cells, thereby accelerating nephropathy progression [8], while simultaneously exerting pro-inflammatory and pro-oxidative effects on cardiac tissues [9]. Recent advancements in glycemic control biomarkers, as reviewed by Krhac et al. [10], highlight gHSA's particular diagnostic utility in heterogeneous diabetic populations, including patients with chronic kidney disease (CKD), where it serves as a valuable adjunctive monitoring parameter. Furthermore, gHSA's regulatory effects on cellular adhesion factors contribute to its systemic impact on cardiovascular health. The dual diagnostic capacity of gHSA as both an inflammatory marker and glycemic indicator enhances its clinical utility. This is particularly evident in its ability to predict mortality risk in diabetic patients undergoing dialysis [11] and its synergistic use with high-sensitivity C-reactive protein (hsCRP) for optimized cardiovascular risk assessment [12]. The integration of gHSA measurements into clinical practice represents a significant advancement in diabetes management and complication prevention strategies.

The nonenzymatic glycation of HSA represents a complex biochemical process involving multiple stages. Initially, reducing sugars covalently bind to protein amino groups, forming reversible Schiff bases. These intermediates subsequently undergo Amadori rearrangement, yielding more stable early glycation products. Over time, these compounds undergo further oxidation, metabolic transformation, and structural rearrangement, ultimately generating advanced glycation end products (AGEs) that play a crucial role in the pathogenesis of various chronic diseases [13]. Szkudlarek et al. demonstrated significant variability in glycation degree measurements across different assay methods, highlighting the influence of sugar specificity and glycation product stages. Their research revealed that while fluorescent assays effectively detect late-stage glycation products, they lack sensitivity for early glycation adducts [14].

Current analytical methodologies for gHSA detection encompass several approaches: High-performance Liquid Chromatography (HPLC) [15], immunoassay techniques [16], thiobarbituric acid colorimetric assays [17], and advanced label-free impedance immunosensors [18]. However, these methods face limitations in clinical application due to procedural complexity, reagent costs, and analytical throughput. The development of rapid, non-invasive gHSA testing methodologies could significantly enhance diabetes screening efficiency and complication monitoring, potentially revolutionizing disease detection capabilities [19]. What's more, different stages of HSA glycation have different effects on the organism, so assays that can analyze different stages of HSA non-enzymatic glycation are also urgently needed, which can be an important guide for diagnosing diseases, etc [20].

As vibrational spectroscopy analysis technology, infrared spectroscopy mainly includes near infrared (NIR) spectroscopy and mid-infrared (MIR) spectroscopy [21]. NIR spectroscopy mainly reflects the multiplicative and additive absorptions of hydrogen-containing functional groups [22]. MIR spectroscopy can reflect the molecular vibration fundamental frequency absorption of most organic compounds [23]. In recent years, with the development of chemometrics technology and new spectrometers [24–26], Infrared spectroscopy is growing rapidly in the field of bio-diagnostics [27–30]. In addition, NIR and MIR spectroscopy have been increasingly applied to the detection of serum

proteins and the diagnosis of diabetes. For example, Wang et al. [31] investigated the feasibility of temperature-controlled NIR spectroscopy for detecting low glucose concentration in serum using mutual factor analysis (MFA), Guo et al. [32] developed a Partial Least Squares (PLS) quantitative model for albumin, globulin, and total protein in serum, and McAvan et al. [33] explored Fourier Transform Infrared (FTIR) spectroscopy for analyzing protein glycosylation. However, most studies were based on a single spectroscopic technique, which limited the ability to quantify gHSA to some extent. Moreover, NIR and MIR spectroscopy each have their own strengths, and the fusion of data from both to quantify gHSA has not yet been attempted.

In this work, the different stages of glycation of HSA were investigated using NIR and MIR spectroscopy combined with principal component analysis (PCA) and continuous wavelet transform (CWT) analysis. In addition, quantitative NIR and MIR models of gHSA were established and the robustness of the model was improved based on the data fusion strategy. This study provided a new research method and perspective on the variations of the glycation process system of HSA and also provided a novel relatively rapid detection method.

2. Materials and methods

2.1. Materials and instruments

HSA (purity $\geq 99.8\%$, SDS-PAGE Sigma-Aldrich, lot No. A21770), PBS buffer (1 $\times$, pH 7.2–7.4, 0.01 M, Solarbio, lot No. 22,080,161,345), D-(+)-Glucose (molecular weight 180.2, Adamas, lot No. C13665375), Fructosamine kit (NBT colorimetric method, Nanjing Jiancheng, lot No. 20,221,026, 20,221,019), 1mm cuvette (21/Q/1, Starna Scientific), 1cm cuvette (Q, Purshee Experiment).

Electronic balance (XSR105, METTLER TOLEDO), NIR spectrometer (Matrix-F, Bruker), FTIR spectrometer (Alpha II, Bruker), Temperature control accessories (Qpod2e, QUANTUMNORTHWEST), Ultraviolet spectra analyzer (Cary 60 UV-Vis, Agilent Technologies).

2.2. Preparation of gHSA

Precision weighing 2.00 g of HSA dissolved in 50 ml of PBS, configured into a concentration of 40 mg/ml of HSA solution, and then fully mixed homogeneously, every 25 ml a share, then divided into two parts. Anhydrous glucose was added to the two portions to configure a system containing 50 mM and 500 mM of glucose, which were mixed well, and each system was divided into an equal number of 12 tubes for a total of 24 samples. The reaction was incubated in the dark at 37 °C for 0 h, 12 h, 1 day, 2 days, 3 days, 4 days, 5 days, 1 week, 2 weeks, 3 weeks, 4 weeks, 5 weeks, and at the end of the incubation, the reaction was quenched by freezing at -20 °C, and the reaction was thawed at 4 °C for the next experimental procedures after 24 h of freezing.

2.3. Acquisition of NIR spectra and MIR spectra

The NIR spectra of the samples were collected using the Matrix-F spectrometer. The measurements were performed in a 1mm cuvette, with a sampling volume of 300 μ l, a wavelength range of 10,000–4000 cm^{-1} , and a spectral resolution of 4 cm^{-1} . The samples were scanned 64 times, and the temperature was controlled at settled room temperature using a temperature control accessory. The air background was subtracted, and three spectra of each sample were collected and averaged.

The MIR spectra of the samples were collected using an ATR transmission accessory of the Alpha II spectrometer. The sampling volume was 10 μ l, and the wavelength range was set to 4000–400 cm^{-1} . The spectrometer had a resolution of 4 cm^{-1} and was scanned 64 times. The temperature was controlled at settled room temperature using a temperature control accessory. The air background was subtracted, and three spectra were collected for each sample and averaged.

2.4. Measurement of HSA glycation levels

The nitro blue tetrazolium (NBT) colorimetric method was used to measure the content of glycated HSA in the samples. The polymeric ketamine substance formed with glucose and HSA undergone a reduction reaction with NBT blue to produce formazan. Fructosamine in dimethylformamide (DMF) was used as a control for colorimetry, enabling measurement of the glycated content. Each sample was tested three times, and the average value was used as the final result.

2.5. Data analysis

We used MATLAB (R2020a, USA) and Origin (2021, OriginLab) to process the spectral data. PCA analysis is a downscaled data processing method that maps multidimensional data features into lower dimensions and extracts information on variables with large explanatory variance. The loading matrix and the score matrix can be obtained by PCA analysis, the loading matrix shows the information of the main feature contributing peaks, and the score matrix represents the weights of the loadings [34]. CWT analysis is an effective method for spectral noise filtering and feature information extraction through the appropriate selection of wavelet functions and wavelet scales [35]. In recent years, CWT has been increasingly used to explore infrared spectral features and reveal molecular interactions [35,36]. However, the extraction of spectral features is affected by different wavelet functions with different vanishing moments. In this study, we used the "sym 4" wavelet function and the "20" wavelet scale to analyse the spectral data.

For the quantitative analysis of gHSA, we developed the PLS quantitative model. First of all, we utilized the sample set partitioning based on joint x-y distance (SPXY) method and divided it into a 3:1 ratio to obtain 18 calibration sets and 6 validation sets. The GA content measured by the NBT colorimetric method was used as primary data. PLS quantitative models were established separately for NIR spectra, MIR spectra, and the fusion of data of NIR and MIR, respectively. The analysis and comparison of the quality of the PLS quantitative models were based on evaluation parameters such as root mean square error of prediction (RMSEP), prediction determination coefficient (R^2_p), relative percent deviation (RPD), as well as root mean square error of calibration (RMSEC) and calibration determination coefficient (R^2_c), among others [37]. RPD, which stands for Relative Prediction Deviation, is the ratio of the standard deviation of the validation set to the root mean square error of the prediction. A model quality is considered reliable when $RPD > 3$. The cross-validation method was used to select the best latent variables (LVs) for participation in PLS modelling. The root-mean-square error of cross-validation (RMSECV) was plotted against the LVs, and the LVs corresponding to the turning points of their descending minimums were selected. Further, we employed standard normalized variate (SNV), mean center (MC), autoscaling (AU), Savitzky-Golay (SG) convolutional derivative, and continuous wavelet transform (CWT) for spectral pre-processing to eliminate spectral noise and enhance spectral resolution. In addition, we selected the characteristic variables using Correlation Coefficient (CC) [38], Monte-Carlo Uninformative Variables Elimination (MC-UVE) [39], Competitive adaptive reweighted sampling (CARS) [40], etc., methods to improve model quality.

3. Results and discussion

3.1. Systematic changes in the glycation process of HSA in the NIR spectral region

Fig. 1(B) showed the contribution share of each principal component (PC), and the first three PCs were able to explain more than 90 % of the information, indicating that the first three PCs could effectively reflect the changes in the system. Fig. 1(C) showed the loadings of PC1, in which the peaks at 6842 cm^{-1} , 5292 cm^{-1} , 4968 cm^{-1} had a high contribution of information, which were mainly attributed to the octave

and merging frequency absorption of water molecules in the system. PC1 scores showed an increasing and then decreasing trend with reaction time going on in Fig. 1(D). In the early stages of the non-enzymatic glycation reaction of HSA, the glucuronide group bound to the free amino group of the protein and dehydrated to form the CONH structure [41]. Late in the reaction, the glycation product was further oxidized and metabolism which consumed water and produced a large number of aldehydes and ketones, reducing the water molecule's information contribution [42]. PC2 mainly represented information from peaks 4613 cm^{-1} and 4383 cm^{-1} (Fig. 1 (E)), which were attributed to $\text{C}=\text{O}$ as well as CH , CH_2 absorptions information in the system, and the overall score was higher during the third week (Fig. 1 (F)). PC3 mainly responded to the information at 4416 cm^{-1} and 4347 cm^{-1} , representing C-O absorptions on the sugar chain as well as information on the combined frequency absorptions of individual CH_2 , and its scores showed a trend of increasing fluctuating potential followed by decreasing and then increasing again, with the lowest scores in the third week in Fig. 1(G and H).

The spectra processed by CWT had more obvious change information in the range of $4200\text{--}4900\text{ cm}^{-1}$, $5600\text{--}6100\text{ cm}^{-1}$, and $6250\text{--}7690\text{ cm}^{-1}$ bands, in which $4200\text{--}4900\text{ cm}^{-1}$ mainly reflected the information of protein amide bands, $5600\text{--}6100\text{ cm}^{-1}$ mainly reflected the information of CH as well as the octave absorption of CH_2 , and the $6250\text{--}7690\text{ cm}^{-1}$ bands mainly reflected the information of the first-octave absorption of O-H in the system. The primary changes in the system as the reaction time increased were the generation of new peaks and the shifting of peak positions.

From Fig. 2, we can find that new peaks were generated at 4322 cm^{-1} and 5764 cm^{-1} , the original peak at 4397 cm^{-1} was moved to 4411 cm^{-1} after 4 days, and the original peak at 6026 cm^{-1} was moved to 6020 cm^{-1} . The peak of 4322 cm^{-1} was attributed to the CH combinatorial absorption and the peak in 5764 cm^{-1} was attributed to the S-H absorption information on proteins, respectively. Thermal breakage of disulfide bonded during non-enzymatic glycation of proteins generates sulfhydryl groups, which enhanced the S-H signal and generates new peaks [43].

The peak at 4397 cm^{-1} was attributed to the combined frequency of individual CH_2 absorptions in the glucose structure, due to the binding of glucose to the protein, which caused a shift in the CH_2 information on the sugar chain, resulting in a shift in the spectral peak [44,45]. The peaks at 6020 cm^{-1} and 6026 cm^{-1} originated from the CH information on the $\text{C}=\text{C}$ bond in the system, and the enolate fructosamine produced by the non-enzymatic glycation process was present with double-bonding information, which changed the position of the spectral peaks due to the presence of enolate and keto interconversions [46]. Amazingly, five new peaks, 4768 cm^{-1} , 5644 cm^{-1} , 5982 cm^{-1} , 7012 cm^{-1} , and 7143 cm^{-1} were generated between 1 week and 3 weeks, and the spectral peaks disappeared after 4 weeks. Combining the results of PCA analysis, it was found that the non-enzymatic glycation of HSA had the highest fructosamine content in the system at 3 weeks, and PC2 represented the information of fructosamine in the system. Therefore, we can speculate that the CWT processing produced five new peaks originating from the characteristic structure of fructosamine in the system. The peak of 4768 cm^{-1} cannot be definitively attributed, but based on previous findings, this peak may represent combined frequency information of C-C vibrations and C-H vibrations on protein side chains, and 5644 cm^{-1} and 5982 cm^{-1} peaks were attributed to the first-order octave absorption of the CH antisymmetric stretch of CH_3 in the side chain and the C-H combinatorial absorption of CH_2 , [47,48], and 7012 cm^{-1} and 7143 cm^{-1} peaks were attributed to the CH combinatorial absorption on the proteins in the system and the O-H hydration layer information on the side chain [49], respectively.

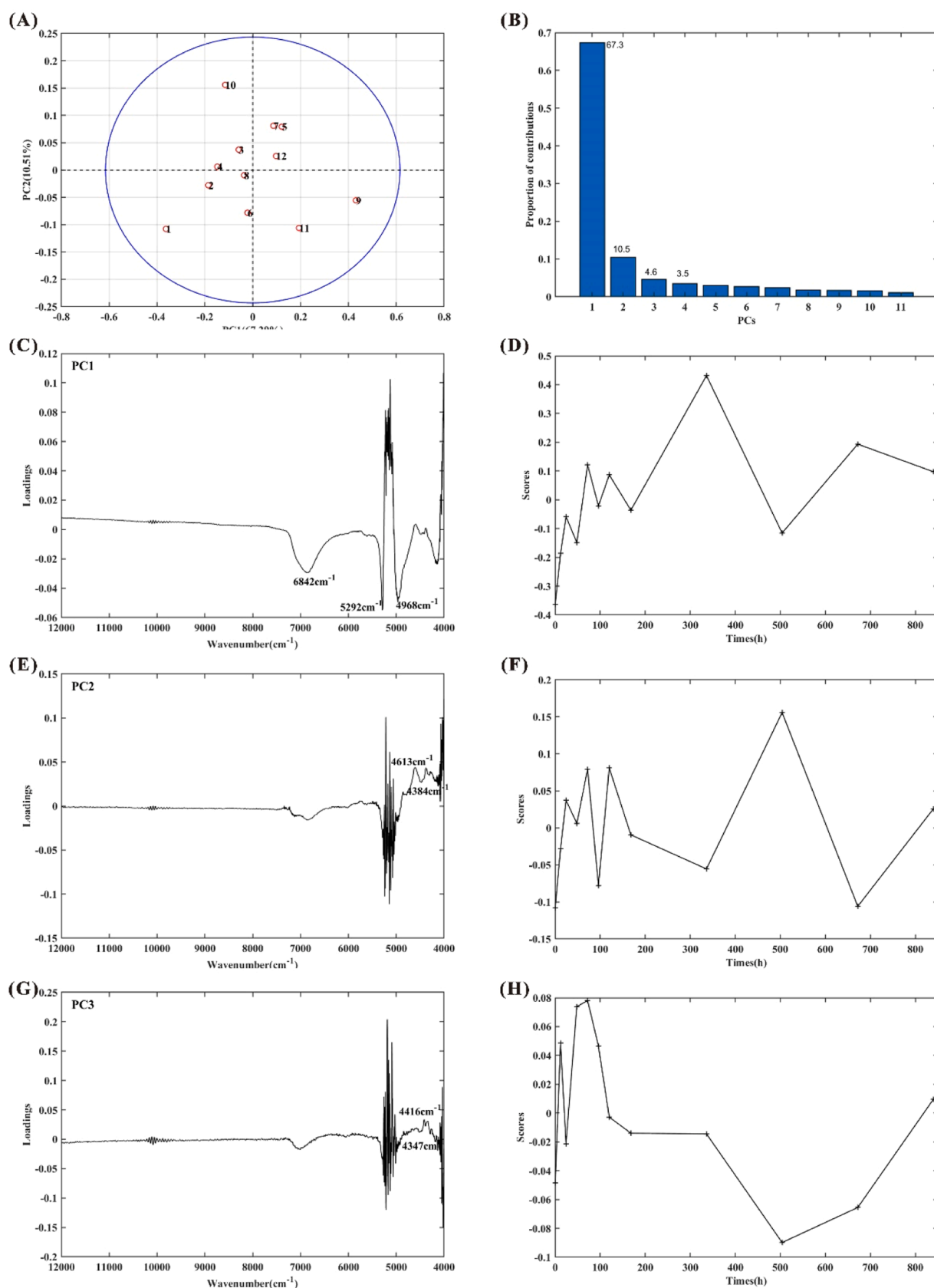


Fig. 1. Plots of NIR spectroscopy PCA analysis of the system at different times of HSA glycation (40 mg/ml). ((A). the plot of the scores of PC1 versus PC2; (B). the percentage of each PC; (C, E, G). the plots of the loadings of PC1, PC2, and PC3, respectively; (D, F, and H). the plots of the scores of PC1, PC2, and PC3 versus the reaction time).

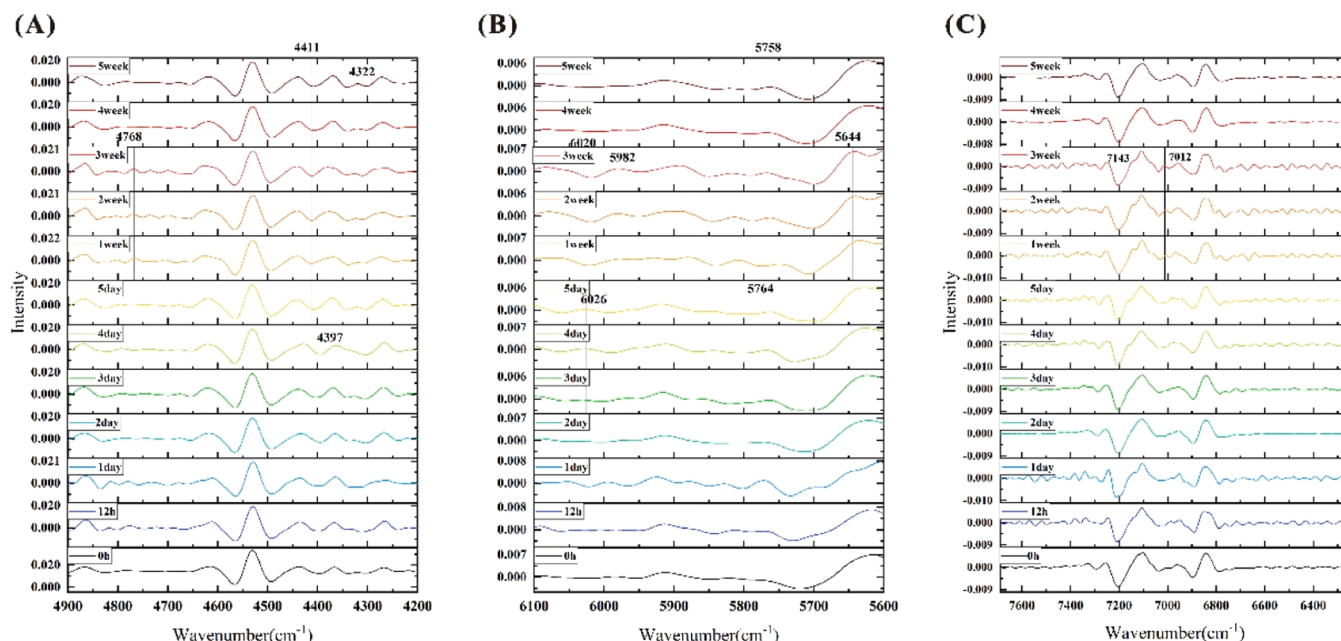


Fig. 2. NIR spectra of HSA ((40 mg/ml)) glycation reaction at different times by CWT treatment ((A) the 4200–4900 cm^{-1} bands; (B) the 5600–6100 cm^{-1} ; (C) the 6250–7690 cm^{-1} bands).

3.2. Systematic changes in the glycation process of HSA in the MIR spectral region

From Fig. 3(B), it can be seen that PC1 explained 77.7 % of the information and PC2 explained 19.2 % of the information. Taken together, they can account for more than 90 % of the total, suggesting that PC1 and PC2 explained most of the information about how the system changed. From Fig. 3(A), it was found that the contribution of PC1 in the system increased as the reaction time increasing, specifically in a phase of 4-day to 5-day, and a phase of 1-week to 5-week. The most important peaks in the PC1 loadings (Fig. 3(C)) are 1685 cm^{-1} , 1517 cm^{-1} , 675 cm^{-1} , whereby 1685 cm^{-1} mainly reflected the vibration absorption of C=O in ketal in the system [50], 1517 cm^{-1} mainly reflected protein tyrosine side chain information [51], and 675 cm^{-1} represented the C-O-C vibration absorption in the protein [52]. The scores of PC1 showed an increasing trend with increasing reaction time in Fig. 3(D), indicating the enhancement of the C=O, C—O—C signal in the system, reflecting the deepening of protein glycation. The main peaks contributing to the PC2 loadings were 1650 cm^{-1} and 1543 cm^{-1} in Fig. 3(E). These peaks represented the vibrational absorption of glucose C=O in the system, the amide I band, and the amide II band information of the protein [33]. Meanwhile, the O-H vibrational absorption information of the water molecule was at 1640 cm^{-1} , and this peak overlapped with the amide I band information [53]. Fig. 3(F) showed a lower level of PC2 scores in the 2-day to 5-day timeframe with an overall downward trend.

The O-H absorption octave frequency of water near 1640 cm^{-1} overlapped with the spectral band near 1650 cm^{-1} of the amide band of proteins, which masked the change information peaks of the system to a certain extent [54]. However, the serious information masking of water can be avoided by analyzing other bands, such as the 600–1500 cm^{-1} band which was the infrared fingerprint region of the compound reflecting the information of deformation vibration of C—H, N—H and stretching vibration of C—O, C—C in the system. Therefore, this band was used as the focus of analysis in this study. According to its spectra by CWT in Fig. 4(A), it can be seen that as the reaction time increasing, new peaks were generated near 600 cm^{-1} , 620 cm^{-1} , 675 cm^{-1} , 710 cm^{-1} , and 1510 cm^{-1} , which corresponded to the peaks in the loadings of PC1. After CWT treatment, the main contributing peaks of PC1 were further separated to generate multiple peaks, but the peaks were still

concentrated near 700 cm^{-1} and 1510 cm^{-1} , which embodied the information of protein glycation products in the system and further proved the deepening of glycation degree. In the 1720–1900 cm^{-1} band as shown in Fig. 4(B), which mainly reflected the C=O vibrational information of the aldehydes and ketones in the system, it can be found that the peak intensities at 1840 cm^{-1} and 1792 cm^{-1} occurred with the increase of reaction time and 1840 cm^{-1} was attributed to the saturated ketone C=O in the system, while 1792 cm^{-1} was attributed to the C—O vibrational information in the system. Pappas et al. [55] found this information in the MIR spectra of milk. This suggested that the sugar reacted with the protein, increasing the C—O bonds in the system.

3.3. Quantitative modelling of gHSA based on NIR spectroscopy

The NIR spectra of the HSA glycation system collected at different sugar concentrations and different incubation times were shown in Fig. 5, with more obvious spectral absorptions near the wavenumbers of 7000 cm^{-1} , 5500 cm^{-1} , 5000 cm^{-1} , and 4600 cm^{-1} . The large absorption peak at 7000 cm^{-1} mainly reflected the first harmonic absorption of the stretching vibration of C—H groups in glucose and protein, as well as the stretching vibration of free O—H groups in water [56]. The small absorption peak at 5500 cm^{-1} mainly reflected the stretching vibration of the CH₂ and CH₃ groups in the system. The large absorption peak at 5000 cm^{-1} mainly reflected the combination frequency absorption of the stretching vibration of free O—H groups and the stretching vibration of N—H [57]. The small absorption peak around 4600 cm^{-1} represented the information on the stretching vibration of C=O groups in protein. The NIR spectra showed that the spectral acquisition process was affected by the instrument and external uncontrollable conditions, resulting in some baseline drift. Therefore, a combination of chemometrics must be used to eliminate the baseline drift and improve the spectral resolution as much as possible. The SNV transform was a pre-processing method that standardized a spectrum by subtracting the mean spectrum and then dividing it by the standard deviation of the data. This normalized the spectrum to a standard normal distribution, thereby eliminating baseline drift caused by scattering.

After other preprocessing techniques had been applied to the SNV data, a quantitative PLS model will be set up. Table 1 indicated that the best PLS quantitative model was established by applying 11 windows

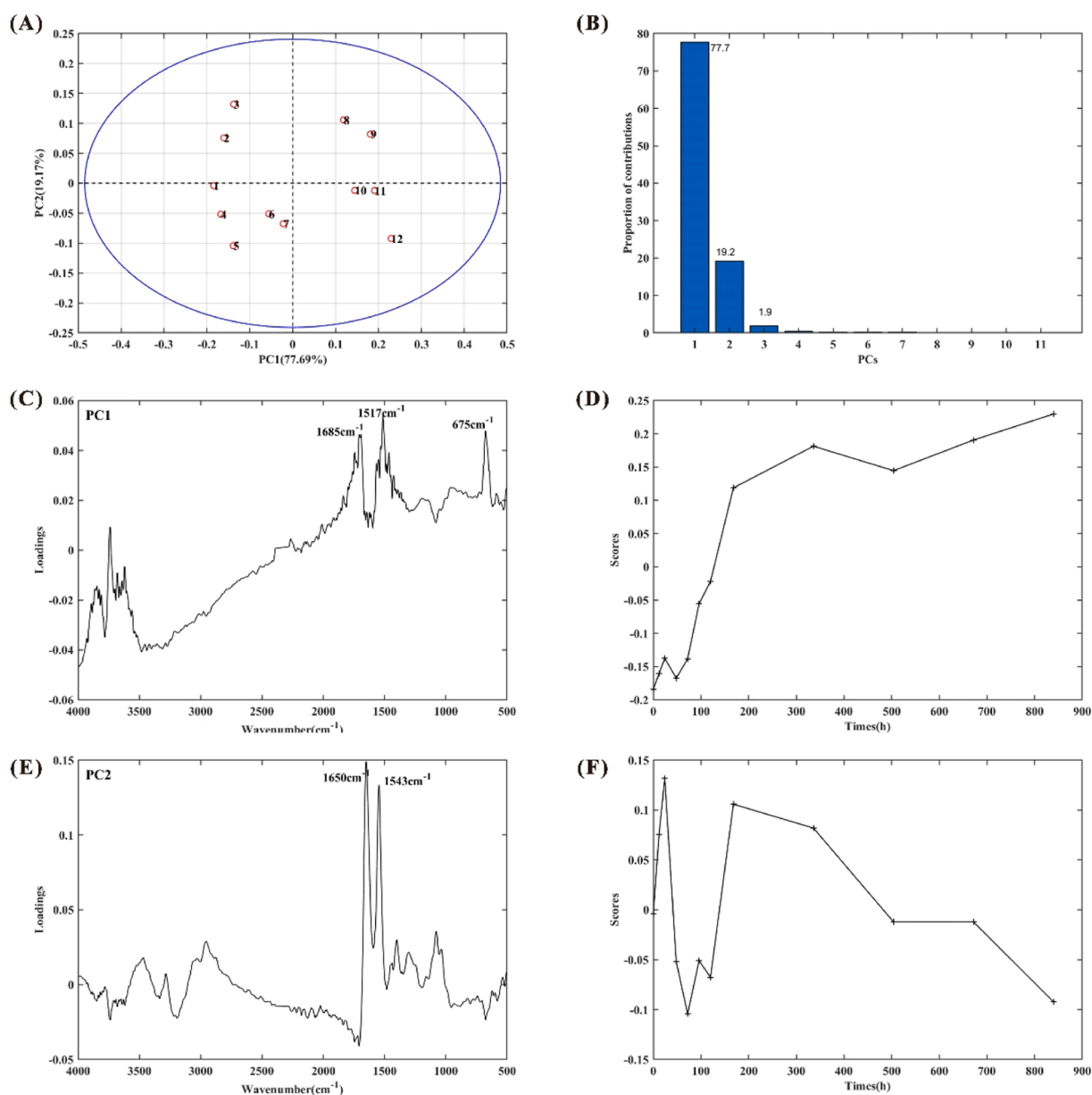


Fig. 3. Plots of MIR spectroscopy PCA analysis of the system at different times of HSA (40 mg/ml) glycation. ((A) the plot of the scores of PC1 versus PC2; (B) the percentage of each PC; (C, E) the plots of the loadings of PC1, PC2, respectively; (D, F) the plots of the scores of PC1, PC2 versus the reaction time)

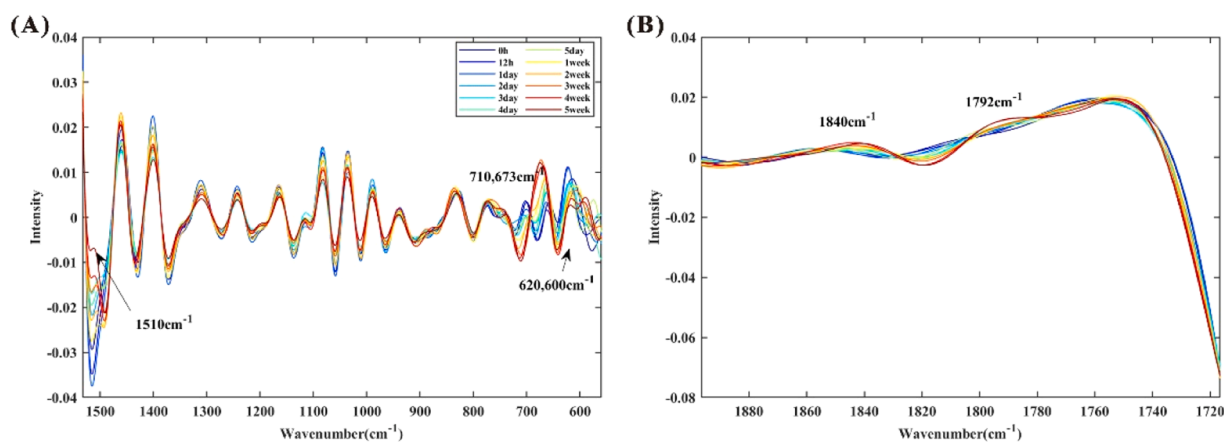


Fig. 4. MIR spectra of HSA (40 mg/ml) glycation reaction at different times by CWT treatment ((A) the 600–1500 cm^{-1} bands; (B) the 1720–1900 cm^{-1} bands).

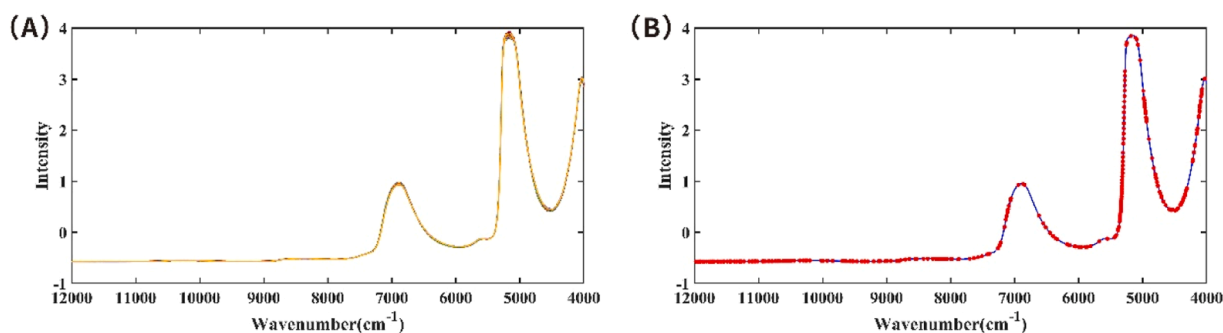


Fig. 5. The NIR spectra after the SNV correction (A) and the characteristic variables were selected by the CC (0.4) method (B). ((B) the blue line was the average NIR spectrum and the red dots were the selected characteristic bands)

Table 1
NIR preprocessing selection criteria.

Methods	LVs	RMSECV (mmol/L)	RMSEC (mmol/L)	RMSEP (mmol/L)	R^2c	R^2p	RPD
SNV+MC	6	4.61	0.44	3.81	0.9955	0.6535	1.40
SNV+AU	6	5.29	0.94	3.73	0.9876	0.6955	1.43
SNV+SG1 +AU	5	3.33	0.11	2.22	0.9997	0.9004	2.40
SNV+SG2+AU	5	3.45	0.13	2.19	0.9996	0.9341	2.43
SNV+CWT (sym2) +AU	7	3.46	0.13	2.13	0.9996	0.9153	2.49
SNV+CWT (sym4) +AU	6	3.67	0.16	2.40	0.9994	0.9015	2.21

Notes: SG1: derivative order = 1; CWT (sym2): wavelet function = sym 2.

and performing the 2nd derivative processing of SG, based on the SNV preprocessing, followed by AU. The RMSEP, R^2p , and RPD of the model were 2.19 mmol/L, 0.9341 and 2.43. However, the RPD of the model was relatively small. Therefore, we conducted variable selection on this preprocessing method to improve the quality of the model by selecting characteristic variables.

Here we mainly examined the three characteristic variable selection methods CC, UVE, and CARS, and selected the best variable selection method based on RMSEP, R^2p , and RPD. Table 2 displayed the results of variable selection. When employing the CC method with a threshold of 0.4, the model achieved its maximum R^2p of 0.9515, with an RMSEP of 2.13 mmol/L and an RPD of 2.5. The number of selected variable points was 565, resulting in an improved model quality. Fig. 5(B) showed the characteristic variables selected by the CC (0.4) method, indicated by the red dots. A total of 565 feature variables were selected across the entire spectral range, avoiding concentration in any particular interval. Based on this, we plotted the RMSECV value against the number of LVs, as shown in Fig. 6(A). The RMSECV was at its minimum with 5 LVs, so we selected this value to establish the CC-PLS quantitative model. As shown in Fig. 6(B), the validation results had a certain amount of error. Therefore, it was necessary to further improve the quality of the model.

Table 2
Parameters for NIR characteristic variable selection method.

Methods	LVs	RMSECV (mmol/L)	RMSEC (mmol/L)	RMSEP (mmol/L)	R^2c	R^2p	RPD	Variable numbers
CC (0.2)	6	2.48	0.05	2.02	0.9999	0.9508	2.60	1877
CC (0.3)	5	2.52	0.08	2.16	0.9998	0.9466	2.46	1079
CC (0.4)	5	1.69	0.09	2.13	0.9998	0.9515	2.50	565
CC (0.5)	2	1.98	1.04	2.51	0.9737	0.9136	0.95	245
MC-UVE (0.9,100)	3	1.46	0.73	2.49	0.9873	0.9045	2.13	581
MC-UVE (0.9,200)	5	1.34	0.16	2.32	0.9994	0.9009	2.29	743
MC-UVE (0.9,300)	5	1.36	0.16	2.36	0.9994	0.9016	2.25	723
MC-UVE (0.9,400)	5	1.46	0.17	2.22	0.9993	0.9140	2.39	673
MC-UVE (0.9,500)	5	1.27	0.17	2.36	0.9993	0.8985	2.25	681
CARS (20)	3	0.50	0.11	3.86	0.9997	0.8207	1.38	50
CARS (40)	4	0.31	0.06	7.34	0.9999	0.5076	0.72	21
CARS (60)	4	0.25	0.05	5.48	0.9999	0.6818	0.97	27
CARS (80)	4	0.35	0.08	4.73	0.9998	0.7658	1.12	27

Notes: CC (0.2): Correlation coefficient threshold = 0.2; MC-UVE (0.9,100): Correlation coefficient threshold = 0.9, Monte Carlo number = 100; CARS (20): Monte Carlo number = 20.

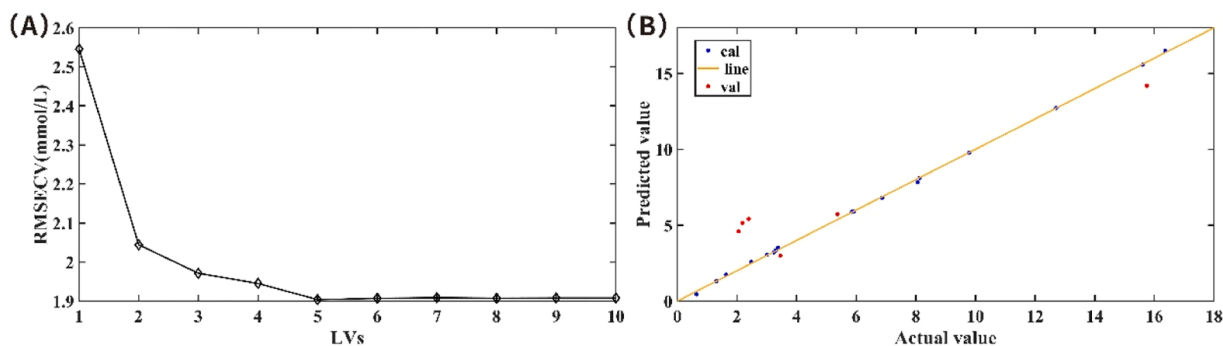


Fig. 6. The optimal number of LVs selected(A) and the CC-PLS quantitative model for NIR spectroscopy (B).

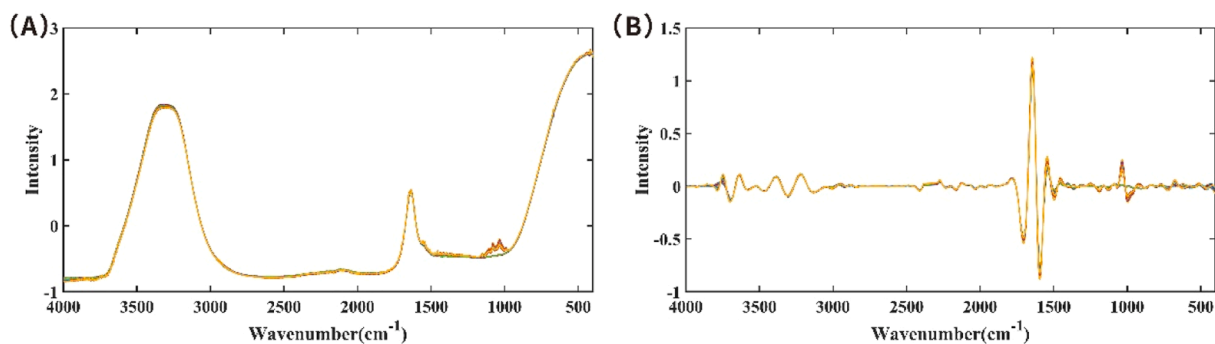


Fig. 7. The MIR spectra after SNV(A) and the MIR spectra by CWT of wavelet functions for sym4 (B).

Table 3

MIR preprocessing selection criteria.

Methods	LVs	RMSECV (mmol/L)	RMSEC (mmol/L)	RMSEP (mmol/L)	R^2c	R^2p	RPD
SNV+MC	6	2.00	0.21	2.61	0.9990	0.8729	2.04
SNV+AU	5	2.26	0.46	2.32	0.9950	0.8934	2.29
SNV+SG1 +AU	5	2.85	0.13	2.26	0.9996	0.8849	2.35
SNV+SG2 +AU	4	3.18	0.06	2.40	0.9999	0.8870	2.21
SNV+CWT (sym2) +AU	3	3.09	0.96	2.24	0.9871	0.8907	2.37
SNV+CWT (sym4) +AU	3	2.71	1.04	1.99	0.9738	0.9345	2.67

of 2.67 when SNV, CWT (sym4), and AU treatment were applied. CWT can be used for denoising and smoothing spectra, extracting feature information. And selecting the appropriate wavelet scale can improve spectral resolution and effectively extract key information. Fig. 7(B) showed the MIR spectrum processed through “sym4” and “30” wavelet scaling, which highlighted significant differences in spectral information.

As shown in Table 4, the full spectra modelling had the best performance, and characteristic variable selection did not improve the

model quality. Therefore, the modelling spectra were selected as the SNV and CWT pre-processed spectra with AU. The best number of LVs, 3, was chosen based on the relationship between RMSECV and LVs, as presented in Fig. 8(A). A PLS quantitative model was established as shown in Fig. 8(B). However, its predictive ability must need to be improved, hence data fusion will be performed next.

Table 4

Parameters for MIR characteristic variable selection method.

Methods	LVs	RMSECV (mmol/L)	RMSEC (mmol/L)	RMSEP (mmol/L)	R^2c	R^2p	RPD	Variable numbers
CC (0.2)	4	2.75	0.45	2.31	0.9952	0.9903	2.30	1335
CC (0.3)	3	2.50	1.08	2.34	0.9721	0.8884	2.27	1140
MC-UVE (0.8,100)	5	1.58	0.59	2.43	0.9918	0.9070	2.19	503
MC-UVE (0.8,200)	3	1.94	1.19	2.41	0.9659	0.8721	2.20	483
MC-UVE (0.8,300)	5	1.62	0.62	2.32	0.9908	0.9051	2.29	497
MC-UVE (0.8,400)	3	1.62	1.07	2.31	0.9722	0.8825	2.30	477
MC-UVE (0.8,500)	5	1.62	1.10	2.31	0.9709	0.8826	2.30	483
CARS (20)	5	0.97	0.11	3.81	0.9997	0.9716	1.40	99
CARS (40)	4	0.61	0.31	4.84	0.9978	0.9896	1.10	19
CARS (60)	8	0.34	0.11	3.80	0.9997	0.9827	1.40	25
CARS (80)	5	0.64	0.12	5.40	0.9996	0.9764	0.98	33
CARS (100)	6	0.54	0.15	4.32	0.9995	0.9759	1.23	40

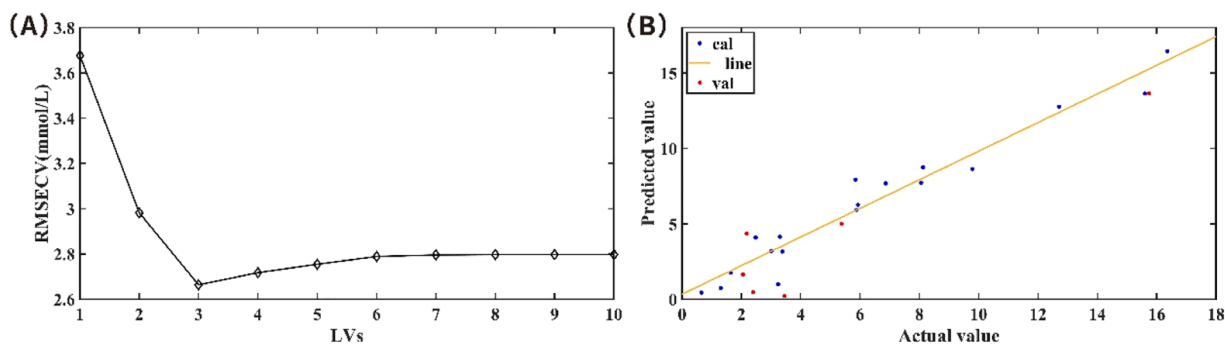


Fig. 8. The optimal number of LVs selected (A) and CC-PLS quantitative model for MIR spectroscopy (B).

3.5. The quantitative PLS model for glycated HSA by the data fusion of NIR and MIR spectroscopy

NIR spectroscopy was better suited for analysing individual components in complex systems, while MIR spectroscopy was fit for structural analysis of materials. The difference between normal HSA and gHSA was little, which was mainly reflected in the modification of protein-free amino acid side chains. Additionally, low-level glycation made the difference even lower. Therefore, the quantitative ability of NIR and MIR spectroscopy analysis techniques was limited when used alone. Infusing NIR and MIR spectroscopy enhanced samples' information and increased the accuracy of quantitative models. Data fusion involved directly combining spectral data matrices to form a new matrix which was pre-processed to enhance model quality.

To prevent the impact caused by the drift of the spectral baseline, we directly merged the NIR spectral data and MIR data for SNV pre-processing. As shown in Fig. 9(A), due to variations in path length and instrument differences, the absorption values for the NIR spectral region were significantly stronger. Therefore, AU was used during the modelling process to ensure that all wavelength variables in the spectra had equal weights. To improve the quality of the quantitative model, the optimal preprocessing method was selected for the fused spectra, as shown in Table 5. The PLS model achieved the best quality when established by SNV, CWT (sym4), and AU treatment, with an RMSEP of 1.88 mmol/L, R^2_p of 0.9249, and RPD of 2.83. Next, based on the pre-processing method, the characteristic variable selection was performed. Table 6 showed the results of the selected characteristic variables using the CC, MC-UVE, and CARS methods. The CC method was found to produce the best modelling results when a threshold of 0.4 was set. The model's RMSEP was 1.59 mmol/L, R^2_p was 0.9524 and RPD was 3.35. In this case, 1445 wavelength points were selected as the feature points, as shown in Fig. 9(B). Based on the relationship between RMSECV and LVs, we selected 4 as the optimal number of LVs for modelling at the inflection point, as shown in Fig. 10(A). Fig. 10(B) showed the optimal PLS

model obtained from fusing the NIR and MIR data, which resulted in an improved predictive capacity of the model.

4. Conclusions

Glycated HSA as a biological marker for disease diagnosis and monitoring has been receiving increasing attention, however, traditional detection methods are time-consuming and laborious. For the study of proteins and molecular interactions, NIR and MIR spectroscopy were increasingly used. Its application in the biomedical field was also increasing and multiple studies showed the feasibility of NIR and MIR in HSA glycation research.

In this paper, we investigated the different levels of glycation of HSA by NIR and MIR spectroscopy combined with PCA and CWT analysis. Based on the NIR, the higher fructosamine content of HSA was found in the third week, and five characteristic peaks 4768 cm^{-1} , 5644 cm^{-1} , 5982 cm^{-1} , 7012 cm^{-1} , 7143 cm^{-1} attributed to fructosamine were extracted and analyzed. What's more, five characteristic peaks 675 cm^{-1} , 1517 cm^{-1} , 1685 cm^{-1} , 1792 cm^{-1} , and 1840 cm^{-1} were found based on the MIR, which were dependent on the time of glycation. These peaks revealed the characteristic ketoaldehyde structure generated and the exposure of protein side-chain tyrosine by the glycation process of HSA. In addition, we also developed PLS quantitative models of gHSA and improved the robustness of the models based on data fusion strategies. The results showed that the PLS quantitative model had the best performance with an RMSEP of 1.59 mmol/L, R^2_p of 0.9524, and RPD of 3.35 after data fusion of NIR and MIR. In general, this study provided a new research method and research perspective on the changes in the glycation process system of HSA, and a novel detection method as well as providing early detection and diagnosis of heterogeneous diabetes mellitus, diabetic complications, and so on. Nevertheless, our study needs to be further developed, and in our next work, we will investigate the effect of sugar concentration as well as the type of sugar on HSA glycation, and explore further the diagnosis of diseases associated with

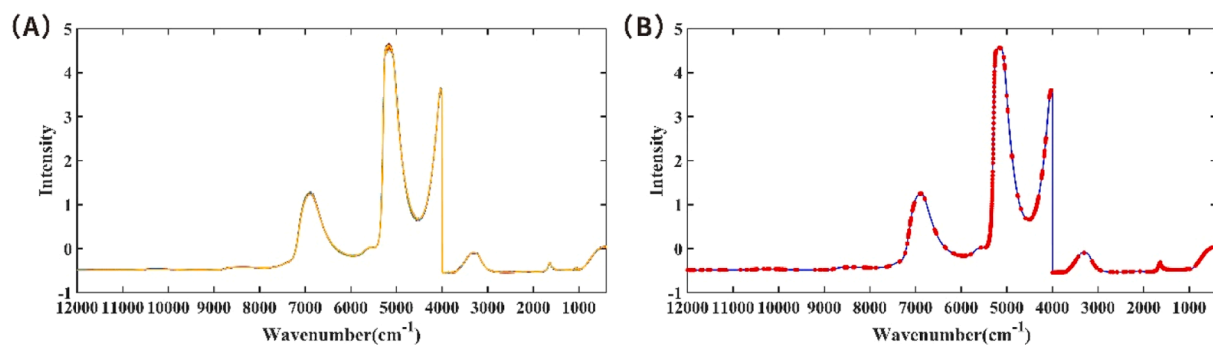


Fig. 9. The data fusion of NIR and MIR after SNV correction (A), and the characteristic variables were selected by CC (0.4) method (B). ((B): the blue line was the average spectrum and the red dots were the selected characteristic bands).

Table 5

Parameters for the preprocessing methods of NIR and MIR data fusion.

Methods	LVs	RMSECV (mmol/L)	RMSEC (mmol/L)	RMSEP (mmol/L)	R^2c	R^2p	RPD
SNV+AU	5	3.16	1.59	2.41	0.9384	0.8845	2.21
SNV+SG1 + AU	4	2.88	0.46	2.12	0.9950	0.9040	2.51
SNV+SG2 + AU	4	3.06	0.33	2.00	0.9974	0.9154	2.66
SNV+CWT (sym2) + AU	4	3.00	0.36	2.00	0.9969	0.9126	2.66
SNV+CWT (sym4) +AU	4	3.05	0.36	1.88	0.9969	0.9249	2.83

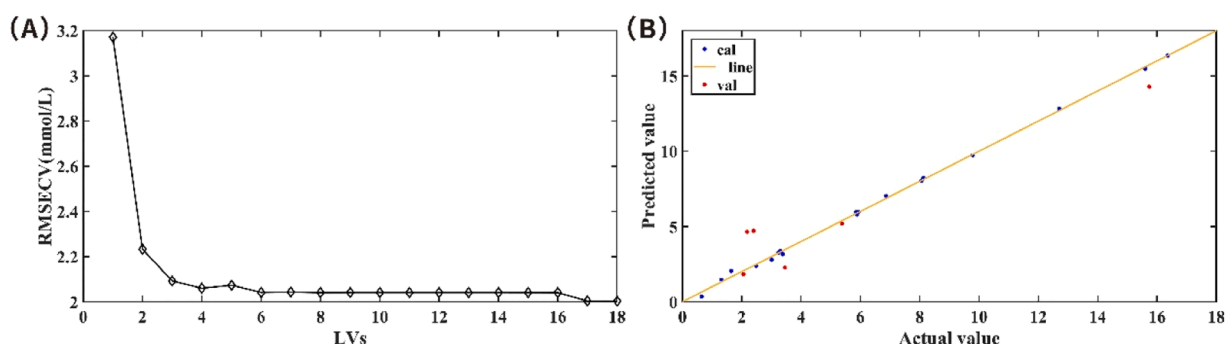
Notes:SG1: SG1: derivative order = 1; CWT (sym2): wavelet function = sym2.

Table 6

Selection of PLS model parameters for NIR and MIR data fusion characteristic variables.

Methods	LVs	RMSECV (mmol/L)	RMSEC (mmol/L)	RMSEP (mmol/L)	R^2c	R^2p	RPD	Variable numbers
CC (0.2)	4	2.42	0.26	1.81	0.9984	0.9333	2.98	3089
CC (0.3)	4	2.24	0.21	1.70	0.9989	0.9443	3.12	2228
CC (0.4)	4	2.06	0.17	1.59	0.9994	0.9524	3.35	1445
CC (0.5)	6	1.67	0.05	1.88	0.9999	0.9236	2.82	831
MC-UVE (0.8,100)	6	1.07	0.08	1.88	0.9999	0.9221	2.83	1230
MC-UVE (0.8,200)	4	1.16	0.29	1.93	0.9980	0.9199	2.76	907
MC-UVE (0.8,300)	3	1.12	0.48	1.79	0.9944	0.9350	2.97	1025
MC-UVE (0.8,400)	6	1.21	0.08	1.88	0.9998	0.9226	2.82	968
MC-UVE (0.8,500)	6	1.22	0.09	1.91	0.9998	0.9200	2.78	1061
CARS (40)	5	0.47	0.10	3.20	0.9998	0.8959	1.66	19
CARS (60)	5	0.37	0.15	4.99	0.9995	0.6176	1.07	20

Notes: CC (0.2): Correlation coefficient threshold = 0.2; MC-UVE (0.8,100): Correlation coefficient threshold = 0.8, Monte Carlo number = 100; CARS (40): Monte Carlo number = 40.

**Fig. 10.** The optimal number of LVs selected(A) and the quantitative PLS model for NIR and MIR data fusion(B).

gHSA in conjunction with aquaphotomics. As for the quantitative model, it is necessary to increase the sample size and complexity and conduct more quantitative research in order to obtain a higher-quality quantitative model.

CRediT authorship contribution statement

Bing Zhao: Writing – original draft. **Hui Zhang:** Writing – review & editing. **Xiuping Liu:** Validation. **Qin Dong:** Conceptualization. **Hengchang Zang:** Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data sets generated and analysed during the current study are available from the corresponding author upon reasonable request.

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